Guanting Liu

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Bio

The Hong Kong Polytechnic University 2026-Now

PhD Student

Guangdong Institute of Intelligence Science and Technology 2025-2026

Research Assistant

Nanjing University of Information Science and Technology 2021-2025

Bachelor of Science

Research Interests

Computational neuroscience, cognitive neuroscience, brain-computer interface, deep learning.

Main Papers

[1] indicates co-first author, * indicates corresponding author.

[1] **Guanting Liu**[1], Ying Yan[1], Jun Cai, Edmond Qi Wu, Shencun Fang*, Adrian David Cheok, & Aiguo Song. (2025). "GCD: Graph contrastive denoising module for GNNs in EEG classification". *Expert Systems with Applications*, 265, 126013. [DOI]

[2] **Guanting Liu***, Ying Yan, Jun Cai, Adrian David Cheok, Edmond Qi Wu, & Aiguo Song. "Rethinking Kalman Filters for Motor Brain-Machine Interface: The Fundamental Limitations and A Perspective Shift". *biorxiv preprint*. [DOI]

[3] **Guanting Liu***, Ying Yan, Jun Cai, Adrian David Cheok, Edmond Qi Wu, & Aiguo Song. "A More Rational and Efficient Kalman Filter Design for Motor Brain-Machine Interfaces". *2025 47th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*. [DOI]

[4] **Guanting Liu***, Ying Yan, Sizhen He, Jun Cai, Adrian David Cheok, Edmond Qi Wu, & Aiguo Song. "A Neuromorphic Approach for Brain-Machine Interface Using Spiking Neural Networks". *2025 47th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*. [DOI]

Other Papers

[5] Ying Yan[1], **Guanting Liu**[1], Jun Cai, Na Liu, Shencun Fang*, Adrian David Cheok, Edmond Qi Wu, Chengcheng Hua, Aiguo Song. (2026). "Residual multi-dimensional Taylor network for epileptic electroencephalography detection". *Engineering Applications of Artificial Intelligence*.

[6] Ying Yan[1], **Guanting Liu**[1], Haoyang Cai, Edmond Qi Wu*, Jun Cai, Adrian David Cheok, Na Liu, Tao Li, & Zhiyong Fan. (2024). "A review of graph theory-based diagnosis of neurological disorders based on EEG and MRI". *Neurocomputing*.

[7] Ying Yan[1], **Guanting Liu**[1], Jun Cai*, Edmond Qi Wu, Adrian David Cheok, Aiguo Song. A Novel Interpretable Multi-Layer Voting Network for Fault Diagnosis. (2026).

[8] Ying Yan*, **Guanting Liu**, Jun Cai, Edmond Qi Wu, Adrian David Cheok, Qiming Sun, Ming Li, & Ying Zhou. (2023). "A Novel ResMTN-Based Methodology for HVAC AHU Fault Diagnosis". *2023 Global Reliability and Prognostics and Health Management Conference (PHM-Hangzhou).*

[9] Haoyang Cai, Ying Yan*, **Guanting Liu**, Jun Cai, Adrian David Cheok, Na Liu, Chengcheng Hua, Jing Lian, Zhiyong Fan, Anqi Chen. (2024) "WKLD-based feature extraction for diagnosis of epilepsy based on EEG". *IEEE Access.*

[10] Ying Yan, Haoran Li, Jun Cai, **Guanting Liu**, Edmond Qi Wu, Hao Wang, Chengcheng Hua, Yaowen Yu*, Aiguo Song. (2026) "A novel correlation-driven cross-term compression polynomial network for classifying motion sickness levels". *Engineering Applications of Artificial Intelligence.*

Research & Projects

Large-scale human-connectome-based task-driven brain modelling 2026-Now

- Based on an MRI structural connectivity scaffold, modeled regional dynamics with firing-rate units and trained the network to perform cognitive tasks.
- Investigated efficient signal propagation and reliable information routing across the whole brain.

Multi-levels cognitive computational modelling 2025-Now

- Modeled biologically constrained models like leaky integrate-and-fire, Hodgkin-Huxley and analyzed their dynamics in cognitive tasks such as working memory and decision making.
- Utilized dynamics tools such as neural manifolds to analyze and compare the computational and neural mechanism.

Whole-brain simulation modelling of zebrafish 2025-2026

- Developed a whole-brain model of zebrafish with more than 70,000 neurons based on real mesoscale connectome, using whole-brain, single-neuron-resolution calcium signals as label for simulation.
- The simulated zebrafish reproduces realistic neural activity dynamics, locomotor behavior outputs, responses to visual stimuli, and neural circuits.

Invasive motor brain-computer interface decoding algorithms designing 2024-2025

- Optimized the classical Kalman filter algorithm in the field of motor control for brain-computer interfaces, improving computational efficiency, accuracy, and interpretability in motor decoding.
- Implemented high-temporal-resolution motor decoding using spiking neural networks.

Non-invasive electroencephalogram signal decoding algorithms designing 2021-2024

- Combined deep learning models such as graph neural networks and convolutional neural networks to achieve EEG decoding for applications including motor imagery and epilepsy detection.
- Investigated feature extraction to derive high-signal-to-noise-ratio features from EEG signals.

Technical Skills

Programming: Python, MATLAB

Languages: Chinese (Native), English (IELTS 7.0), Cantonese (Beginner)